

Simple Shell (Command Runner) Project

Features:

- Accept commands from the user
- Create a child process using fork()
- Execute commands using execvp()
- Parent waits for child using wait()
- Handle invalid commands gracefully
- Exit using exit command

Source Code (C)

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];
    while (1) {
        printf("SimpleShell> ");
        fgets(input, MAX_INPUT, stdin);
        input[strcspn(input, "\n")] = 0;
        if (strcmp(input, "exit") == 0) break;
        int i = 0;
        args[i] = strtok(input, " ");
        while (args[i] != NULL && i < MAX_ARGS - 1) {
            i++;
            args[i] = strtok(NULL, " ");
        }
        pid_t pid = fork();
        if (pid == 0) {
            execvp(args[0], args);
            perror("Command execution failed");
            exit(1);
        } else {
            wait(NULL);
        }
    }
    return 0;
}
```

Compile and Run

```
gcc simple_shell.c -o shell
./shell
```

Sample Run

```
SimpleShell> pwd
/home/user
```

```
SimpleShell> ls
file1.txt
file2.txt
```

```
SimpleShell> hello
Command execution failed: No such file or directory
```

```
SimpleShell> exit
Exiting shell...
```

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main.cpp

Share

Run

Output

Clear

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <string.h>
4 #include <unistd.h>
5 #include <sys/wait.h>
6
7 #define MAX_INPUT 1024
8 #define MAX_ARGS 64
9
10 int main() {
11     char input[MAX_INPUT];
12     char *args[MAX_ARGS];
13
14     while (1) {
15         printf("SimpleShell> ");
16
17         if (fgets(input, MAX_INPUT, stdin) == NULL)
18             break;
19
20         input[strcspn(input, "\n")] = 0;
21
22         if (strcmp(input, "exit") == 0)
23             break;
24
25         int i = 0;
26         args[i] = strtok(input, " ");
27
28         while (args[i] != NULL && i < MAX_ARGS - 1) {
29             ...
30         }
31     }
32 }
```

SimpleShell> |

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main.cpp

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```
17 if (fgets(input, sizeof input, stdin) == NULL)
18     break;
19
20 input[strcspn(input, "\n")] = 0;
21
22 if (strcmp(input, "exit") == 0)
23     break;
24
25 int i = 0;
26 args[i] = strtok(input, " ");
27
28 while (args[i] != NULL && i < MAX_ARGS - 1) {
29     i++;
30     args[i] = strtok(NULL, " ");
31 }
32
33 pid_t pid = fork();
34
35 if (pid == 0) {
36     execvp(args[0], args);
37     perror("Command execution failed");
38     exit(1);
39 } else {
40     wait(NULL);
41 }
42 }
43
44 return 0;
45 }
```

SimpleShell> |